

Markscheme

Mathematics: analysis and approaches
Practice question bank

41. (a)

Consider the equation $\log_3(x + 2) - \log_3(x - 1) = 2$, where $x > 1$.

Solve the equation for x , giving your answer as an exact fraction or decimal.

combine using the quotient law: $\log_3\left(\frac{x + 2}{x - 1}\right) = 2$ (M1)

$$\frac{x + 2}{x - 1} = 9$$
 A1

$$x + 2 = 9x - 9$$
 (M1)

$$x = 1.375$$
 A1

[4 marks]